Econ 240B: Lecture 3

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This lecture is about asymptotics.

1 Sample Average Asymptotics

The sample average given a population size n is given by

$$\bar{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i$$

where $Y_i \sim iid\left(\mu, \sigma^2\right)$. The claim is that for any $c > 0$,

$$CI_n\left(\mu\right) = \left[\bar{Y}_n - c\frac{\sigma}{\sqrt{n}}, \bar{Y}_n + c\frac{\sigma}{\sqrt{n}}\right]$$

satisfies

$$\begin{aligned} P\left(\mu \in CI\left(\mu\right)\right) = ... = P\left(-c \leq \frac{\sqrt{n}\left(\bar{Y}_n - \mu\right)}{\sigma} \leq c\right) \\ \to_{n\to\infty} \Phi\left(c\right) - \Phi\left(-c\right) \\ = 1 - 2\Phi\left(-c\right) \end{aligned}$$

where $\Phi\left(\cdot\right)$ is the standard normal cdf.

1.1 Justification

For every $t \in \mathbb{R}$,

$$P\left(\frac{\sqrt{n}\left(\bar{Y}_n - \mu\right)}{\sigma} \leq t\right) \to_{n\to\infty} \Phi\left(t\right)$$

which we can rewrite using different notation as

$$\frac{\sqrt{n}\left(\bar{Y}_n - \mu\right)}{\sigma} \to_d N\left(0, 1\right)$$

Definition 1. Let $\{T_n\}$ be a sequence of statistics (random vectors) of the same dimension. Then T_n converges in distribution to T, $T_n \to_d T$, iff

$$P\left(T_n \leq t\right) \to_{n\to\infty} P\left(T \leq t\right)$$

for any continuity points t of $P\left(T \leq \cdot\right)$.

2 Central Limit Theorem

Theorem 2. *Lindeberg-Levy Central Limit Theorem (CLT). If $Y_i \sim iid\left(\mu, \sigma^2\right)$, then*

$$\sqrt{n}\left(\bar{Y}_n - \mu\right) \to_d N\left(0, \sigma^2\right)$$

which can be restated as

$$\frac{\sqrt{n}\left(\bar{Y}_n - \mu\right)}{\sigma} \to_d N\left(0, 1\right)$$

therefore,

$$P\left(\frac{\sqrt{n}\left(\bar{Y}_n - \mu\right)}{\sigma} \leq t\right) \to_{n\to\infty} \Phi\left(t\right), \forall t$$

Convergence in distribution involves convergence of CDF working, but working with CDF's is tedious.

Theorem 3. *Helly-Brey. For a statistics,*

$$T_n \to_d T$$

iff

$$\mathbb{E}\left[g\left(T_n\right)\right] \to_{n\to\infty} \mathbb{E}\left[g\left(T\right)\right]$$

for every bounded and continuous $g\left(\cdot\right)$.

This theorem is used to prove CLT and explains why we need the carve out for discontinuity.
If $T_n \to_d T$ and if $P\left(T_n \leq \cdot\right)$ is continuous then

$$P\left(T_n \leq t\right) \to_{n\to\infty} P\left(T \leq t\right), (\forall t) \tag{1}$$
$$P\left(T_n \leq t\right) \to_{n\to\infty} P\left(T \leq t\right), (\forall t) \tag{2}$$

where (1) and (2) are equal when the CDF is continuous and represents point-wise convergence. However, this is also true for uniform convergence:

$$\sup_t \left|P\left(T_n \leq t\right) - P\left(T \leq t\right)\right| \to_{n\to\infty} 0$$

2.1 Question

Let

$$\hat{CI}_n\left(\mu\right) = \left[\bar{Y}_n - c\frac{\hat{\sigma}_n}{\sqrt{n}}, \bar{Y}_n + c\frac{\hat{\sigma}_n}{\sqrt{n}}\right]$$

where $\hat{\sigma}_n^2$ is an estimator of σ. Under what conditions do we have

$$P\left(\mu \in \hat{CI}\left(\mu\right)\right) \to_{n\to\infty} 1 - 2\left(-c\right), \forall c$$

2.2 Algebra

The coverage probability for our CI estimator is

$$\begin{aligned} P\left(\mu \in \hat{CI}\left(\mu\right)\right) &= P\left(\bar{Y}_n - c\frac{\hat{\sigma}_n}{n} \leq \mu \leq \bar{Y}_n + c\frac{\hat{\sigma}_n}{\sqrt{n}}\right) \\ &= P\left(c \leq \frac{\sqrt{n}\left(\bar{Y}_n - \mu\right)}{\hat{\sigma}_n} \leq c\right) \end{aligned}$$

therefore, it suffices to assume that

$$\frac{\sqrt{n}\left(\bar{Y}_n - \mu\right)}{\hat{\sigma}_n} \to_d N(0,1)$$

This rewrite is called studentization – everything other than the parameter of interest can be calculated from the data.

2.3 Answer

The answer to question 2.1 is trivial if we used an oracle estimator an $\hat{\sigma}_n^2 = \sigma^2$. However, the conditions are still met if

$$\hat{\sigma}_n^2 \to_d \sigma^2$$

or

$$\hat{\sigma}_n^2 \to_p \sigma^2$$

In fact, these are the same statements and the next remainder will show both why these conditions are necessary and why they are equivalent.

2.4 Claim

Let

$$\begin{aligned} U_n &= \sqrt{n}\left(\bar{Y}_n - \mu\right) \\ V_n &= \hat{\sigma}_n^2 \end{aligned}$$

and let

$$\begin{aligned} U &\sim N\left(0, \sigma^2\right) \\ V &= \sigma^2 \end{aligned}$$

If

$$(U_n, V_n) \to (U, V)$$

the,

$$\frac{\sqrt{n}\left(\bar{Y}_n - \mu\right)}{\hat{\sigma}_n} = \frac{U_n}{\sqrt{V_n}} \to_d \frac{U}{\sqrt{V}} \sim \frac{N\left(0, \sigma^2\right)}{\sqrt{\sigma^2}} \sim N(0,1)$$

To prove this, we will rely on the following theorem,

Theorem 4. *Continuous Mapping Theorem (CMT). If $T_n \to_d T$ and if g is a function with $P(T \in C_g) = 1$, where C_g contains the points of g, then*

$$g(T_n) \to_d g(T)$$

Now consider the following application, for any $u \in \mathbb{R}$, $v > 0$, let

$$g(u, v) = \frac{u}{\sqrt{v}}$$

then,

- $g(\cdot)$ is continuous on $\mathbb{R} \times \mathbb{R}_{++}$

- $P(U \in \mathbb{R}, v > 0) = 1$

Therefore, we can apply our theorem and get

$$\frac{U_n}{\sqrt{V_n}} \to_d \frac{U}{\sqrt{V}} \sim N(0,1)$$

if

$$(U_n, V_n) \to_d (U, V)$$

with

$$U \sim N\left(0, \sigma^2\right)$$
$$V = \sigma^2$$

Fact 5. *If $U_n \to_d U$ and $V_n \to_d V$ insofar as V is degenerate, then*

$$(U_n, V_n) \to_d (U, V)$$

Recall some of the following heuristics:

- In general, marginals do not pin down joint distributions
- The joint distributions of U and V is determined by the marginals only if $U \perp V$
- But, if V is degenerate, then $U \perp V$

Therefore, if $\hat{\sigma}_n^2 \to_d \sigma^2$, then

$$\frac{\sqrt{n}\left(\bar{Y} - \mu\right)}{\hat{\sigma}_n^2} \to_d N(0,1)$$

Definition 6. Let $\{T_n\}$ be a sequence of random vectors. T_n converges to T in probability,

$$T_n \to_p T$$

iff

$$P\left(||T_n - T|| \leq \delta\right) \to_{n \to \infty} 1$$

This is just a generalization of convergence. Non-randomness will imply normal convergence, but this definition accounts for randomness.

We also say that $\hat{\sigma}_n^2$ is a consistent estimator of σ^2 iff $\hat{\sigma}_n^2 \to_p \sigma^2$.

Thus, if we have a statistic $T_n \to_p T$, then we must also have $T_n \to_d T$. These results rely on the continuity assumptions of CDFs made earlier. The converse is not generally true unless T is degenerate. Therefore, if

$$\hat{\sigma}_n^2 \to_p \sigma^2$$

then

$$\frac{\sqrt{n}\left(\bar{Y} - \mu\right)}{\hat{\sigma}_n^2} \to_d N(0,1)$$

3 Bottom line

To summarize, if we have asymptotic normality

$$\sqrt{n}\left(\bar{Y}_n - \mu\right) \to_d N\left(0, \sigma^2\right)$$

and if have consistency

$$\hat{\sigma}_n^2 \to_p \sigma^2$$

then

$$\frac{\sqrt{n}\left(\bar{Y}_n - \mu\right)}{\hat{\sigma}_n^2} \to_d N(0,1)$$